BSc in Data Science for Responsible Business

Program Overview and Course – Academic Year 2026–2027



we are makers

CENTRALE LYON

em lyon business school



Overview of the programme

Key points of the programme

The BSc in Data Science for Responsible Business is a joint program between **emlyon business school** (ranked among the top 10 business schools in Europe, Financial Times 2024) and **École Centrale de Lyon** (member of the prestigious Groupe des Écoles Centrale, consistently ranked among France's leading engineering institutions). Together, they have created an elite, highly selective undergraduate program designed for ambitious students aiming to become the next generation of data scientists, capable of combining technical excellence with business acumen and an ethical understanding of AI and data-driven technologies.

🎓 Program Architecture (Years 1–4)

The BSc in Data Science for Responsible Business is an elite and selective four-year program designed for highly capable students with strong quantitative, computational, and analytical backgrounds. The program provides a comprehensive education that combines technical mastery with management and responsible leadership.

- **Year 1: Foundations in mathematics, statistics, economics, computer science, and management.**
- **Year 2: Transition to advanced, graduate-level courses in mathematics, statistics, programming, optimization, and AI.**
- **Year 3: International mobility in the Fall term (students study abroad, while the program welcomes incoming exchange students in the Fall, Spring, or both terms). In the Spring term, students specialize in advanced data science applications and applied projects.**
- **Year 4: Capstone year with electives, data scientist, research or entrepreneurship tracks, and a full-semester professional internship**

The program meets corporate and market needs for data scientists who combine technical and analytical expertise with business understanding and a strong awareness of the human, ethical, and legal dimensions of AI.

Organisation of courses

Overview of the programme

🌎 **Incoming Exchange Admissions & Academic Calendar Requirements**

Applicants from partner universities seeking to complete an exchange semester or full exchange year within the BSc in Data Science for Responsible Business must be enrolled in a computer science or data science or related program. Their applications reviewed by the Academic Direction of the BSc, which will determine their admission status. Incoming exchange students must comply with the academic calendar of the BSc program.

Academic Calendar

Fall Term: The Fall term begins in the last week of August or the first week of September and concludes in the second or third week of December.
Spring Term: The Spring term begins in the first or second week of January and ends in the first or second week of May.

Language of instruction : English

🎓 Credit allocation

Incoming students can enroll in one of the following packs each totaling 30ECTS credits.

FALL PACK 🍁

- Numerical Methods for Optimization (5 ECTS)
- Statistical Learning (5 ECTS)
- CSR and AI for Society (5 ECTS)
- AI Culture (5 ECTS)
- Machine Learning in Practice (5 ECTS)
- Natural Language Processing and Understanding (5 ECTS)

SPRING PACK 1 🌷

- Big Data (5 ECTS)
- Deep Learning (5 ECTS)
- Agile Software Development (5 ECTS)
- Big Data Analytics for Business (5 ECTS)
- Marketing & Managerial Finance (5 ECTS)
- Natural Language Processing and Understanding (5 ECTS)

SPRING PACK 2 🌸

- Deep Learning Projects (5 ECTS)
- Organizational Behavior & Strategy (5 ECTS)
- Responsible Algorithmics (5 ECTS)
- Corporate Law (5 ECTS)
- AI Responsible Mission – Entrepreneurship (5 ECTS)

⚠️ **Once the semester starts, changes or cancellations of courses will not be permitted.**

FALL SCHEDULE

Monday	Tuesday	Wednesday	Thursday	Friday
@Centrale			@emlyon	

Welcome Days : August 27th

Fall term start : Monday August 31st , 8am

Fall term break : from October 26th to October 30th

Fall term end: December 18th

SPRING SCHEDULE

Monday	Tuesday	Wednesday	Thursday	Friday
@Centrale		@emlyon		

Spring term start : Monday January 4th , 8am

Spring term break : from February 22th to February 26nd

Spring term end : May 24th



BSc course catalog

FALL PACK

- Numerical Methods for Optimization (5 ECTS)
- Statistical Learning (5 ECTS)
- CSR and AI for Society (5 ECTS)
- AI Culture (5 ECTS)
- Machine Learning in Practice (5 ECTS)
- Natural Language Processing and Understanding (5 ECTS)

Numerical Methods for Optimization

ECTS: 5 credits

Course level: Intermediate

Course description

This course provides a rigorous foundation in numerical methods and optimization, the mathematical frameworks underlying modern data science and machine learning. Beginning with core theoretical principles and advancing to practical implementation, students learn to formulate and solve complex optimization problems commonly encountered in business and analytical domains.

The course integrates both continuous and discrete perspectives, covering methods such as gradient descent, Lagrange multipliers, and graph-based algorithms including Dijkstra's method for network analysis. Special care is devoted to implementing these techniques in Python to address real-world computational challenges, equipping students with the analytical and practical skills necessary for advanced study in areas such as machine learning and deep learning.

Assessment tools: midterm exam, final exam.

Prerequisite: Statistics, Linear Algebra, Multivariate Calculus

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Define and interpret core concepts in optimization theory and graph theory. 2. Describe the fundamental principles underlying numerical methods for optimization. 3. Identify and characterize the essential components of linear programming formulations. 4. Explain the significance of vector space norms and their role in numerical analysis. 5. Depict how graphs are used to model and analyze network-structured problems.
6.1.2 Knowledge Acquired During the Course	1. Explain the mechanics, convergence behavior, and limitations of iterative optimization algorithms. 2. Distinguish between discrete and continuous optimization frameworks and their respective applications. 3. Identify ill-conditioned mathematical problems and analyze their implications for machine learning and numerical stability. 4. Compare the efficiency, accuracy, and trade-offs among various numerical methods. 5. Describe the mathematical foundations of algorithms such as PageRank, Spectral Clustering, and related graph-based techniques.
6.1.3 Competencies Developed During the Course	1. Implement numerical algorithms in Python. 2. Select the appropriate optimization strategy for a given problem. 3. Evaluate the effectiveness and efficiency of a numerical solution. 4. Collaborate effectively in a team to solve technical programming assignments. 5. Design a computational solution to a business problem using numerical methods.

Statistical Learning

ECTS: 5 credits

Course level: Advanced

Course description

This course provides a comprehensive introduction to the core concepts and methods of statistical learning with a focus on both theoretical understanding and practical application for data-driven decision making. Topics covered include linear regression and classification, model selection, regularization techniques, (Ridge, Lasso), resampling methods (cross-validation, bootstrap), tree-based methods, and an introduction to support vector machines and unsupervised learning. The course emphasizes the bias-variance trade-off, model interpretability, and the ethical application of predictive models in a business context.

Assessment tools: Midterm, Final exam

Prerequisite: Statistics, Multivariate Calculus, Linear Algebra

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. The Bias-Variance Trade-off 2. Supervised vs. Unsupervised Learning 3. Model Assessment and Selection 4. Linear Methods for Regression and Classification 5. Tree-Based and Ensemble Methods
6.1.2 Knowledge Acquired During the Course	1. Mathematical foundations of key statistical learning algorithms 2. Principles and applications of regularization and resampling 3. Criteria for evaluating model performance and generalizability 4. Strengths and weaknesses of different learning approaches 5. The importance of interpretability and ethics in machine learning
6.1.3 Competencies Developed During the Course	1. Select the optimal model using cross-validation and hyperparameter tuning 2. Critically interpret and communicate the results of a predictive model 3. Design and execute a complete machine learning pipeline for a business problem

CSR and AI for Society

ECTS: 5 credits

Course level: Intermediate

This course is made of two parts:

1) System Thinking and Sustainability 2, and

2) Responsible Commitment

CSR and AI for Society – System Thinking and Sustainability 2

Course level: Intermediate

Course description

This course builds on “systems thinking and sustainability”, together designed to develop and manage strategies and policies in an increasing complex world with mounting global and societal challenges. This second course focuses on using system dynamics tools for modeling tools, including simulation and integration of data. Critical application issues of learning, developing insights, high-leverage policies, implementation, and confidence building will be discussed.

Assessment tools: Class participation, Several group projects (written and oral reports)

Prerequisite: System Thinking and Sustainability 1 or equivalent

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	Systems thinking tools, processes, and perspectives: In this course students will be introduced to the basic principles and fundamental tools of system dynamics modeling, with focus on organizational or societal problems within their broader environment in terms of the structures and policies that create dynamics and regulate performance
6.1.2 Knowledge Acquired During the Course	1. a variety of systems thinking tools including causal loop diagramming and especially simulation to explore feedback and dynamic behaviour of systems. 2. how to explore, analyze, and critique existing models, learn the opportunities and challenges of working with data, and get to understand knowledge to be created and its limits. 3. examine different real-world model applications designed for studying long-term side effects of decisions, systematically explore new strategies, and develop understanding of complex systems.
6.1.3 Competencies Developed During the Course	1. Analyze, present, and discuss dynamically complex problems using systems thinking tools (eg problem formulation, causal mapping, analysis of dynamics) considering key stakeholders of the problems. 2. Understand the essence of the modelling process, including fundamentals of dynamic model formulation, computational modeling, challenging the boundary, model testing, analysis through simulation, and data-based analysis 3. Understand how to use models and data to look for high-leverage sustainability solutions and climate action, from the individual to the global level, considering the long-term, collective behaviour, and side-effects.

CSR and AI for Society

CSR and AI for Society – Responsible Commitment

Course level: Intermediate

Course description

The course aims at developing reflexive skills concerning the challenges of developing or adopting AI in a responsible way. It also aims at providing the foundations for an adequate understanding and management of Corporate Social Responsibility. Finally, the course should also help students to develop a capacity to take an ethical standpoint in decision making.

Assessment tools: Individual self-assessement, Several group projects (written and oral reports)

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. History of AI 2. Ethics 3. Extended responsibilities for people and the planet 4. Anthropology
6.1.2 Knowledge Acquired During the Course	1. Understand ethics in a data driven world 2. Understand managers and engineers' responsibilities in a data driven work environment 3. Discover new perspectives for economic and political development 4. Discover and deepen citizen's participations in political life an decision making
6.1.3 Competencies Developed During the Course	1. Use data and AI responsibly in managerial and engineer's practices 2. Contribute to new ethical economic and political development 3. Leverage technical data mastering and ethical awareness to contribute to new business and political development

AI Culture

ECTS: 5 credits

Course level: Beginner

This course is made of three parts:

1) History and Perspectives of AI,

2) Open Data and Privacy, and

3)AI Ethical Concerns

AI Culture – History and Perspectives of AI

Course level: Beginner

Course description
Learn the history of AI, and reflect about its development perspectives and consequences for business and society. Understand the stakes of data and open data for business and governance. Discover the fundamentals of ethical concerns around the development of AI, and the law and regulation stakes (in particular in the EU).

Assessment tools: Final written exam, Group project (written and oral reports)

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. History of sciences 2. History of AI
6.1.2 Knowledge Acquired During the Course	1. The three phases in the history and development of AI 2. Understand scientific perspectives of AI
6.1.3 Competencies Developed During the Course	1. Envision how modern trends in AI are rooted in the history of sciences. 2. Think critically about AI and its perspectives 3. Make projections about scientific, business, and human perspectives of AI.

AI Culture

AI Culture – Open Data and Privacy

Course level: Beginner

Course description

This course aims at giving to students some knowledge and thoughs to develop strategy options in terms of data management as well as critical views about data sources, and data management more largely.

The course explores various topics which include: 1) Open data concept in a data strategy, 2) Privacy, and 3) introduction to data governance

Assessment tools: Class participation, Case studies, MCQS

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Data strategy 2. Open data 3. Data regulation 4. Privacy 5. Data governance
6.1.2 Knowledge Acquired During the Course	1. Identify data strategy steps to implement 2. Identify the impacts of data regulation on such data strategy 3. Identify the limits regarding data regulation and privacy concerns
6.1.3 Competencies Developed During the Course	1. Design a simple data strategy 2. Apply data regulations to this data strategy 3. Consider ethical concerns about data usage and privacy

AI Culture

AI Culture – AI Ethical Concerns

Course level: Beginner

Course description

This course reflects on what is intelligence and what is AI. It discusses the impacts of AI and the perspectives if opens. After a review of the state of the art in AI, the course studies the genuine cognitive impacts of technology, in particular digital and AI powered media, The course addresses the question of what human intelligence is, how it differs from AI, as well as what is creativity.

Assessment tools: In class debates, In class presentation of various works (books, articles, reviews…)

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Artificial Intelligence 2. Human intelligence and cognition 3. Psychology and psychoanalysis views on human cognition 4. The human attention and its biases
6.1.2 Knowledge Acquired During the Course	1. Understand applications of AI to business and society 2. Understand what human intelligence is and how it differs from AI 3. The models and topics of clinical psychology and psychoanalysis 4. What is human creativity 5. How digital media use human attention
6.1.3 Competencies Developed During the Course	1. Reflect on human intelligence and cognition 2. Identify one's own cognitive biases 3. Identify one's own attention biases 4. Acquire a critical understanding of how AI is used in digital media 5. Identify cognitive impacts of AI and learn how to counter them

Machine Learning in Practice

Machine Learning in Practice

ECTS: 5 credits

Course level: Advanced

Course description

A hands-on and practical approach to machine learning, based on past kaggle competitions and with a focus on model interpretation. Technical aspects range from data preprocessing to fine-tuning advanced regression and classification models with random forests and gradient boosting models and handling large datasets.

Assessment tools: Individual and group projects (code, written report, pitch)

Prerequisite: Python programing, algorithmics, multivariate calculus, linear algebra

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Machine Learning 2. Ensemble Models 3. Random Forests 4. Gradient Boosting
6.1.2 Knowledge Acquired During the Course	1. Principles of data processing 2. Principles of handling large datasets 3. Regression with ensemble models 4. Classification with ensemble models
6.1.3 Competencies Developed During the Course	1. Collect and process data for a business need. 2. Build and fine-tune ensemble learning models for regression and classification tasks 3. Handle large dataset (preprocessing and modeling) 4. Interpret ensemble models to answer business questions

Natural Language Processing and Understanding

ECTS: 5 credits

Course level: Advanced (graduate level)

Course description

This course provides a comprehensive introduction to natural language processing and natural language understanding, and their practical usage through applied projects. It covers the classical foundations of NLP: text preprocessing, tokenization, stemming and lemmatization, text representation (bag-of-words, n-grams, vector representations, etc.), topic modeling models (LDA, and modern transformer-based models), and modern transformer-based LLM architectures for NLU.

Assessment tools: Weekly projects to implement in practice what has been covered in class, Several more in-depth projects to perform complex NLP/NLU tasks

Prerequisite: Statistics, Linear Algebra, Multivariate Calculus

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. NLP and NLU 2. Latent Dirichlet Allocation 3. Transformers 4. Large Language Models
6.1.2 Knowledge Acquired During the Course	1. Understand NLP and NLU foundations 2. Understand LDA, BERTopic, VADER and similar models for topic modelling 3. Understand transformers and LLMs
6.1.3 Competencies Developed During the Course	1. Preprocess textual data 2. Perform classical NLP and modern NLU tasks on a corpus of documents 3. Perform topic modeling with LDA models, BERTopic, VADER, etc. 4. Fine-tune an LLM to a specific context 5. Know how to build a RAG on a specific context



BSc course catalog

SPRING PACK 1 🌷

- Big Data (5 ECTS)
- Deep Learning (5 ECTS)
- Agile Software Development (5 ECTS)
- Big Data Analytics for Business (5 ECTS)
- Marketing & Managerial Finance (5 ECTS)
- Natural Language Processing and Understanding (5 ECTS)

Big data

ECTS: 5 credits

Course level: Advanced (graduate level)

Course description

This course provides a foundational introduction to the core concepts, architectures, and technologies involved in the analysis and processing of large-scale data. Emphasis is placed on understanding the structure and function of modern Big Data ecosystems, as well as on developing the technical and analytical skills necessary to apply these systems effectively. Through a combination of lectures, guided laboratory sessions, and collaborative projects, students will gain hands-on experience with representative technologies such as Hadoop, Linked Open Data, and MongoDB. By the end of the course, students will be able to analyze large and complex datasets, implement data-processing workflows, and evaluate Big Data solutions within practical and organizational contexts.

Assessment tools: various projects, midterm exam, final project.

Prerequisite: SQL, Python programming

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Big Data and Open Data 2. Linked Open Data 3. MapReduce paradigm 4. Document-oriented databases 5. Cloud computing and AWS certifications
6.1.2 Knowledge Acquired During the Course	1. Explain the fundamental principles and architectures of Big Data systems and distributed computing environments. 2. Describe key technologies used for large-scale data storage, processing, and analysis. 3. Apply basic data processing workflows using Big Data frameworks and tools. 4. Discuss the role of cloud platforms in enabling scalable data solutions. 5. Evaluate different data management and processing approaches in terms of efficiency and scalability.
6.1.3 Competencies Developed During the Course	1. Implement Big Data frameworks to solve real-world data analysis problems. 2. Deploy and manage data solutions using cloud platforms, particularly AWS. 3. Organize and query data within distributed and document-oriented environments. 4. Analyze and interpret large datasets using appropriate Big Data tools and methodologies. 5. Assess emerging technologies and trends in Big Data and Cloud Computing for continuous professional development.

Deep learning

ECTS: 5 credits

Course level: Advanced (graduate level)

Course description

This course introduces Deep Learning, covering topics from multi-layer neural networks to advanced architectures such as Convolutional Neural Networks (CNNs), Transformers, Generative Adversarial Networks (GANs), and Diffusion Models. It combines theoretical foundations with hands-on practice using the PyTorch library. Application use cases include Computer Vision, Natural Language Processing, and Business-oriented problems.

Assessment tools: various projects, midterm exam, final project.

Prerequisite: Statistical learning, Python programming

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Deep Learning architectures and optimization principles 2. Convolutional Neural Networks (CNNs) for visual representation learning 3. Transformers and Large Language Models (LLMs) for sequential and contextual learning 4. Generative models (Autoencoders, VAEs, GANs, Diffusion Models) 5. Retrieval-Augmented Generation (RAG) systems and hybrid AI architectures
6.1.2 Knowledge Acquired During the Course	1. Understanding and implementing machine learning models (linear, logistic regression, neural networks) 2. Explaining and applying deep learning training methodology (backpropagation, regularization, optimization) 3. Evaluating model performance and interpreting results in computer vision and NLP tasks 4. Applying PyTorch and Hugging Face libraries for model development and experimentation 5. Integrating advanced generative and retrieval-augmented models into real-world applications
6.1.3 Competencies Developed During the Course	1. Building, training, and fine-tuning deep learning models for various applications 2. Designing and implementing advanced neural network architectures (CNNs, Transformers, GANs, Diffusion) 3. Using PyTorch and Hugging Face tools to develop and deploy AI solutions 4. Critically analyzing model performance and improving architectures based on empirical results 5. Developing end-to-end AI pipelines, including RAG systems, for real-world business and research problems

Agile Software Development

ECTS: 5 credits

Course level: Beginner

Course description

This 45-hour course introduces students to Agile software development through a practical and hands-on learning experience. It combines project management, teamwork, low-level programming in C, Cython modules, version control with Git/GitHub, software testing, and DevOps concepts using GitHub Actions for continuous integration. Students participate in workshops where they use Visual Studio or VS Code to write and debug C programs, create small performance-oriented Cython modules, collaborate through GitHub, and build simple CI workflows with GitHub Actions. All these skills come together in a final team project that includes Agile planning, C/Cython development, testing, documentation, and a final demo.

Assessment tools: Final team project in SCRUM, Workshops and labs, Quizzes and MCQS, Class participation.

Prerequisite: Python programming

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Learn and apply Agile teamwork practices in a real project setting. 2. Use Git and GitHub professionally for collaboration (branches, commits, pull requests). 3. Write, structure, and debug low-level C programs using Visual Studio or VS Code. 4. Accelerate Python code using Cython by creating small C extensions. 5. Understand and apply DevOps basics using GitHub Actions for continuous integration and automated testing.
6.1.2 Knowledge Acquired During the Course	1. Understanding how to organise and manage a small Agile project: creating a backlog, writing user stories, planning sprints, assigning tasks, and conducting reviews. 2. Practical experience using Git and GitHub: branching, clean commits, resolving conflicts, opening pull requests, reviewing code, and tracking tasks with GitHub Issues. 3. Essential knowledge of low-level C programming: compilation, functions, modular code, pointers, memory allocation, arrays, segmentation faults, and debugging tools. 4. Ability to write Cython modules: creating .pyx files, compiling extensions, and interfacing Python with C to improve performance. 5. Understanding of core software testing techniques: unit tests, expected outputs, regression testing, and debugging strategies to ensure code reliability. 6. Awareness of DevOps practices through GitHub Actions: building simple CI pipelines, automating tests, and interpreting CI feedback directly in GitHub.
6.1.3 Competencies Developed During the Course	1. Team collaboration through Agile ceremonies (planning, stand-ups, reviews, retro-spectives). 2. Professional use of Git/GitHub workflows (branches, pull requests, code reviews). 3. Ability to build, debug, and improve low-level C programs in a modern IDE. 4. Skill in integrating C code with Python using Cython for performance. 5. Ability to design and run meaningful tests to validate program behavior. 6. Practical understanding of continuous integration with GitHub Actions.

Big Data Analytics for Business

💡 **ECTS: 5 credits**

Course level: Beginner and Intermediate

This course is made of three parts:

1) **Introduction to Digital Marketing**,

2) **Power BI**, and

3) **AWS Cloud Computing**

Big Data Analytics for Business – Introduction to Digital Marketing

Course level: Beginner

Course description

In today's marketing world, understanding digital tools and **data-driven strategies** is essential. This course will help students move beyond intuition and learn how to make **evidence-based marketing decisions.** Through a **hands-on simulation**, students will experience how data guides media resource allocation and discover **the power of A/B testing** to optimize performance. By the end, they will understand how data transforms digital marketing from guesswork into strategic decision-making.

Assessment tools: Business game, MCQS, Final exam

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. A/B testing 2. Conversion funnel 3. POEM digital marketing ecosystem (paid, owned, earned media) 4. 30+ digital KPI 5. The online consumer journey 6. The storytelling models
6.1.2 Knowledge Acquired During the Course	1. the principles of media attribution and how different channels contribute to marketing outcomes. 2. the purpose of A/B testing for evaluating strategies. 3. any budget allocation logic to optimize marketing investments. 4. data-driven decision-making frameworks linking analytics to marketing choices.
6.1.3 Competencies Developed During the Course	1. Design and manage a digital marketing strategy aligned with brand objectives 2. Analyze and optimize the online customer journey 3. Use data and analytics to support decision-making and improve marketing performance 4. Apply A/B testing methods to allocate media resources efficiently 5. balance creativity, technology, and ethics in digital marketing practices

Big Data Analytics for Business – Power BI

Course level: Intermediate

Course description

This course introduces the reference business analyst tool: Microsoft power BI. It is intended for mastering business intelligence. Business Intelligence Analysts gather, analyze, and visualize data to support business decisions. They create reports using Power BI to drive strategy and improve organizational performance.

The course covers data preparation (harnessing, loading, transformations, etc), data analysis (including statistical analysis), data modeling (clustering, and supervised learning), and data viz, data dashboarding, and data reporting for decision-making.

Assessment tools: Case studies based on Microsoft PowerBI certification, in class studies, presentations (oral and written)

Prerequisite: databases (principally SQL), statistics, dataviz

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. business intelligence 2. data driven decision-making 3. data analysis & statistical analysis
6.1.2 Knowledge Acquired During the Course	1. data pipelines with powerBI 2. data handling and modeling with powerBI 3. data driven decision-making with powerBI
6.1.3 Competencies Developed During the Course	1. prepare, load, and transform data in powerBI 2. data analysis and statistical analysis with powerBI 3. data modeling with powerBI 4. dataviz, dashboarding, reporting with powerBI 5. data driven decision-making with powerBI to answer business questions

Big Data Analytics for Business – AWS Cloud Computing

Course level: Beginner

Course description

The course is a hands-on preparation for the AWS cloud practitioner certification. Students will learn the working environment of AWS cloud, and its various solutions for data storage, data management and modeling, and cloud computing. As the Amazon Academy puts it: “In today’s digital-first world, understanding the cloud has become an essential skill for a wide range of careers. Whether you're stepping into cloud for the first time, leading IT teams, or expanding your technical expertise, knowledge of cloud is essential to innovate and scale. Amazon Web Services (AWS) continues to be a key player in the cloud space, and we’ve recently updated our AWS Cloud Practitioner Essentials course to provide you with the most relevant, up-to-date knowledge. We’ll take you through the fundamentals of AWS services, pricing, security, monitoring, and architecture.”

Assessment tools: in class studies and exercises, final AWS certification

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Cloud environment 2. Cloud computing 3. AWS data architectures 4. AWS data management
6.1.2 Knowledge Acquired During the Course	Foundational, high-level understanding of AWS Cloud, services, and terminology: 1. AWS cloud concepts 2. Security and compliance in the AWS Cloud 3. Core AWS services 4. Economics of the AWS Cloud
6.1.3 Competencies Developed During the Course	1. Explain the value of the AWS Cloud. 2. Understand and explain the AWS shared responsibility model. 3. Understand security best practices. 4. Understand AWS Cloud costs, economics, and billing practices. 5. Describe and position the core AWS services, including compute, network, database, and storage services. 6. Identify AWS services for common use cases.

Marketing and Managerial Finance

💡 **ECTS: 5 credits**

Course level: Beginner

This course is made of two parts:

1) Discovering Marketing, and

2) Managerial Finance & Cost Management

Marketing and Managerial Finance – Discovering Marketing

Course level: Beginner

Course description
The course is based on lecturers and sessions including a very complete business game, designed for beginners and intermediate participants willing to practice marketing in team.

The very interactive, fun, realistic platform is one of the most famous marketing Simulation in the world and the Teacher is certified to play it (INSEAD has designed the game, INSEAD Is the #1 MBA in Europe).

At the end of the course, the participant will know and use the vocabulary associated with the basic principles of marketing and understand the key stages of strategic and operational marketing to be able to identify trends, analyze the competition, segment a market, define a priority target and build the essential elements of a distinctive offer.

Assessment tools: Business game, group presentation/pitch, individual presentation/pitch

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Strategic marketing: segmentation, targeting, positioning 2. Operational marketing declined with the 4 P's 3. ROI: Return On Investment 4. BCG Matrix, a critical perspective 5. Mc Kinsey Matrix, a critical perspective
6.1.2 Knowledge Acquired During the Course	1. the marketing vocabulary 2. the steps to construct BCG and Mc Kinsey Matrix 3. the way to calculate the ROI of a marketing project 4. the manner to use data in a marketing project
6.1.3 Competencies Developed During the Course	1. perfectly build, use and criticize the BCG and Mc Kinsey matrix in a stragegic report 2. read and analyse economic data on a market 3. build, read and analyse a basic Cost Accounting Table 4. self-analyse a deep understanding of a group project in a limited time

Marketing and Managerial Finance – Managerial Finance & Cost Management

Course level: Beginner

Course description

The course is designed to help future managers acquire a sound understanding of the role of managerial finance and cost accounting in organizations. Students are expected to develop a basic theoretical and practical grasp of managerial accounting tools, techniques and documents useful to analyze complex situations such as anticipating future outcomes of present decisions. This course introduces fundamental managerial finance concepts providing students with a solid understanding of technical issues in the design and functioning of budgets and reporting as well as knowledge of cost accounting methods and Cost-Volume-Profit analysis. Participants will be made familiar with forecasts and variance analysis.

Assessment tools: Final exam, MCQS

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Budgeting 2. Cost-Volume-Profit analysis 3. Marginal costing 4. Variance analysis
6.1.2 Knowledge Acquired During the Course	1. Participate to budgeting and reporting procedures 2. Use appropriate managerial finance tools in decisions making 3. Implement sound reasonings grounded on management accounting concepts
6.1.3 Competencies Developed During the Course	1. use methods and techniques of forecasting 2. understand and apply formulas used in budget control (e.g. variances) 3. identify the merits and limitations of managerial finance

Natural Language Processing and Understanding

ECTS: 5 credits

Course level: Advanced (graduate level)

Course description

A comprehensive course which covers natural language processing and natural language understanding, and their practical usage through applied projects. It covers the classical foundations of NLP : text preprocessing, tokenization, stemming and lemmatization, text representation (bag-of-words, n-grams, vector representations, etc.), topic modeling models (LDA, and modern transformer-based models), and modern transformer-based LLM architectures for NLU.

Assessment tools: Weekly projects to implement in practice what has been covered in class, Several more in-depth projects to perform complex NLP/NLU tasks

Prerequisite: Statistics, Linear Algebra, Multivariate Calculus

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. NLP and NLU 2. Latent Dirichlet Allocation 3. Transformers 4. Large Language Models
6.1.2 Knowledge Acquired During the Course	1. Understand NLP and NLU foundations 2. Understand LDA, BERTopic, VADER and similar models for topic modelling 3. Understand transformers and LLMs
6.1.3 Competencies Developed During the Course	1. Preprocess textual data 2. Perform classical NLP and modern NLU tasks on a corpus of documents 3. Perform topic modeling with LDA models, BERTopic, VADER, etc. 4. Fine-tune an LLM to a specific context 5. Know how to build a RAG on a specific context



BSc course catalog

SPRING PACK 2 🌸

- Deep Learning Projects (5 ECTS)
- Organizational Behavior & Strategy (5 ECTS)
- Responsible Algorithmics (5 ECTS)
- Corporate Law (5 ECTS)
- AI Responsible Mission – Entrepreneurship (5 ECTS)

Deep learning

ECTS: 5 credits

Course level: Advanced (graduate level)

Course description

This course introduces Deep Learning, covering topics from multi-layer neural networks to advanced architectures such as Convolutional Neural Networks (CNNs), Transformers, Generative Adversarial Networks (GANs), and Diffusion Models. It combines theoretical foundations with hands-on practice using the PyTorch library. Application use cases include Computer Vision, Natural Language Processing, and Business-oriented problems.

Assessment tools: various projects, midterm exam, final project.

Prerequisite: Statistical learning, Python programming

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Deep Learning architectures and optimization principles 2. Convolutional Neural Networks (CNNs) for visual representation learning 3. Transformers and Large Language Models (LLMs) for sequential and contextual learning 4. Generative models (Autoencoders, VAEs, GANs, Diffusion Models) 5. Retrieval-Augmented Generation (RAG) systems and hybrid AI architectures
6.1.2 Knowledge Acquired During the Course	1. Understanding and implementing machine learning models (linear, logistic regression, neural networks) 2. Explaining and applying deep learning training methodology (backpropagation, regularization, optimization) 3. Evaluating model performance and interpreting results in computer vision and NLP tasks 4. Applying PyTorch and Hugging Face libraries for model development and experimentation 5. Integrating advanced generative and retrieval-augmented models into real-world applications
6.1.3 Competencies Developed During the Course	1. Building, training, and fine-tuning deep learning models for various applications 2. Designing and implementing advanced neural network architectures (CNNs, Transformers, GANs, Diffusion) 3. Using PyTorch and Hugging Face tools to develop and deploy AI solutions 4. Critically analyzing model performance and improving architectures based on empirical results 5. Developing end-to-end AI pipelines, including RAG systems, for real-world business and research problems

Organizational Behavior and Strategy

ECTS: 5 credits

Course level: Intermediate

This course is made of three parts:

1) Organizational Behavior, and

2) Strategy

Organizational Behavior and Strategy – Organizational Behavior

Course level: Intermediate

Course description

OB is an interdisciplinary field of study concerned with the human dimensions of work organisations. Whether you are a naturally talented people person and a team player, or whether you prefer numbers crunching or systems design, organizations are unthinkable without people, and you will need to make sense of their behaviours once you join a workplace. In any career, you will be working for, and with, people who have expectations of you - understanding their and your own behaviour and how it is affected by different contexts, situations, and people is a valuable ability that you need to master.

This course will introduce you to both organizational behaviour and organizational studies. We will cover key topics from each of the three levels of analysis: individual (perception, personality, and motivation), group (teams and team dynamics), and structures (leadership, power, control and resistance). Adapting our analytical focus in this way encourages us to think critically and creatively about the relationship between individuals and structures; about the various ways in which people shape organizations and are, in turn, shaped by organizations.

The main objectives of this course are:

- to provide an understanding of key theoretical contributions to organizational behaviour and organization studies
- to enable students to use these theories to critically analyse organizational practices, and to begin using examples of these to critically reflect on organizational theories
- to acquire and develop fundamental skills in research, critical analysis, use of evidence and development of insight in the field of OB

Assessment tools: reports (written and oral), field study.

Prerequisite: Introduction to Management, Business Game

Organizational Behavior and Strategy

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Major approaches to the study of behaviour at work 2. Individual and group decision-making processes 3. Personality theories and models 4. Motivation theories 5. Leadership theories 6. Emotions and conflict at work
6.1.2 Knowledge Acquired During the Course	1. The effects of group dynamics on individuals' attitudes and behaviours 2. the most common psychological phenomena impacting individual performance within the context of the organisation 3. the main approaches to work motivation the main theories on personality and how it impacts work behaviour 4. the major approaches to leadership in organizations
6.1.3 Competencies Developed During the Course	1. diagnose organizational behaviour issues using core OB concepts and evidence 2. develop and justify solutions to people-management problems drawing on OB theories and research 3. collaborate effectively in diverse teams, demonstrating communication, conflict management, and basic leadership skills 4. demonstrate self-awareness and ethical responsibility regarding one's own behaviour at work and its impact on others 5. communicate OB analyses and recommendations professionally, in both written and oral form

Organizational Behavior and Strategy – Strategy

Course level: Intermediate

Course description

This course is an introduction to business strategy, and its process. It covers strategic analysis, both internal and external, generic corporate strategies, business models and alternative strategizing principles, as well as organization and leadership for digital strategies.

Assessment tools: interactive lectures participation, case research and discussion, oral presentations.

Prerequisite: Introduction to Management, Business Game

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. corporate strategy 2. strategic analysis 3. business model 4. digital strategy
6.1.2 Knowledge Acquired During the Course	1. understanding of key framework of competitive business strategy 2. understanding of business models and alternative strategizing principles
6.1.3 Competencies Developed During the Course	1. develop reasoning based on mainstream and alternative concepts of business strategy 2. know how to apply key frameworks of business strategies to generic context 3. know how to apply key frameworks of business strategies to the context of digitized industries

Responsible Algorithmics

ECTS: 5 credits

Course level: Intermediate and Advanced

This course is made of three parts:

1) Ethical Design,

2) Algorithmic Risks and Audit, and

3)Introduction to Cybersecurity

Responsible Algorithmics – Algorithmic Risks and Audit

Course level: Intermediate

Course description

This course focuses on systematically identifying, assessing, and mitigating risks in algorithmic systems. Students will learn comprehensive auditing methodologies to evaluate models for fairness, security, performance, and compliance. The curriculum covers risk assessment frameworks, audit protocols, and governance mechanisms to ensure algorithmic accountability and trustworthiness throughout the system lifecycle.

Assessment tools

Prerequisite: Algorithmics

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Techniques for pre- and post-deployment impact assessments 2. Stress-testing and red-teaming algorithmic systems 3. Documentation standards for audit trails and compliance reporting 4. Continuous monitoring approaches for model performance and drift
6.1.2 Knowledge Acquired During the Course	1. Techniques for pre- and post-deployment impact assessments 2. Methods for stress-testing and red-teaming algorithmic systems 3. Documentation standards for audit trails and compliance reporting 4. Continuous monitoring approaches for model performance and drift 5. Sector-specific regulatory requirements and compliance protocols
6.1.3 Competencies Developed During the Course	1. Design and execute comprehensive algorithmic audit plans 2. Conduct risk analysis and develop mitigation strategies 3. Create governance frameworks and accountability mechanisms 4. Communicate audit findings to technical and executive audiences 5. Develop monitoring systems for ongoing risk management

Responsible Algorithmics – Ethical Design

Course level: Advanced

Course description

This course equips students to design and advocate for ethically sound algorithms. It focuses on translating ethical principles into practical value by examining the real-world risks and impacts of algorithmic bias, opacity, and unfairness. The curriculum covers ethical frameworks, fairness metrics, and explainable AI (XAI) through the lenses of stakeholder trust, regulatory compliance, and overall system robustness.

Assessment tools : Tests, Final exam

Prerequisite: Algorithmics

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Types and sources of algorithmic bias (e.g., historical, representation, measurement). 2. Theories of fairness. 3. Principles of Explainable AI (XAI) and Interpretability. 4. Transparency, accountability, and redress in intelligent systems.
6.1.2 Knowledge Acquired During the Course	1. Key technical definitions and metrics for fairness. 2. Methods for detecting and quantifying bias in datasets and model outputs. 3. Techniques for model explainability (e.g., SHAP, LIME) and their interpretation. 4. Strategies for integrating ethical checkpoints into a project development lifecycle.
6.1.3 Competencies Developed During the Course	1. Conduct an ethical risk assessment for a data science project. 2. Apply fairness metrics and explainability tools to evaluate a model. 3. Propose design modifications to mitigate identified biases. 4. Document a model's limitations and ethical considerations clearly. 5. Discuss the ethical implications of an algorithmic system with technical and non-technical stakeholders.

Responsible Algorithmics – Cybersecurity

Course level: Intermediate

Course description

This course provides a comprehensive overview of cybersecurity principles tailored for the data science lifecycle. Students will learn to identify, assess, and mitigate security risks in data collection, storage, processing, and model deployment. The curriculum covers secure coding, data privacy laws, adversarial machine learning, and cloud security, combining theory with hands-on labs to build secure and trustworthy data-driven systems.

Assessment tools : Quizzes, Final exam

Prerequisite: Programming, Computer Architecture

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Confidentiality, integrity, availability in a data context. 2. Threat modeling for data pipelines and ML systems. 3. Core principles of cryptography and Identity & Access Management. 4. Data privacy frameworks and regulations.
6.1.2 Knowledge Acquired During the Course	1. Secure data handling practices (encryption, anonymization). 2. Cloud security configuration for data storage and computation. 3. Common vulnerabilities in ML models and data science applications. 4. Defensive techniques against adversarial attacks. 5. Key requirements of major data privacy regulations.
6.1.3 Competencies Developed During the Course	1. Conduct a security risk assessment for a data science project. 2. Implement secure coding and data handling protocols. 3. Competence in hardening a cloud environment for data workloads. 4. Analyze and enhance model robustness against attacks. 5. Communicate security risks and requirements to stakeholders.

Corporate Law

ECTS: 5 credits

Course level: Intermediate

Course description

The aim of this course is to acquire general knowledge of the founding principles of law, contract law and to identify the particularities of the rules concerning commercial agreements and intellectual property.

The course also deals with the application of legal issues in the digital ecosystem, in particular data protection and AI.

The analysis and understanding will be made through lectures, case studies, group works and practical examples.

Assessment tools: MCQS, presentation and pitch.

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Introduction to law 2. Main principles of contract law 3. Commercial agreement (sales, agency, distribution and franchise) 4. Intellectual property (copyright, patents, trademark and design) 5. Data protection 6. AI and main legal issue
6.1.2 Knowledge Acquired During the Course	1. The main rules concerning data protection in the EU 2. The main legal issues of AI 3. The nature of a contract, the rules of law applicable to commercial agreements 4. The fundamental rules concerning intellectual property 5. The legal implications and risks regarding intellectual property rights
6.1.3 Competencies Developed During the Course	1. recognize and use basic legal concepts, prioritize information 2. Identify a legal problem and choose between various forms of legal solutions (from a manager's perspective), being aware of all potential risks involve.

AI Responsible Mission – Entrepreneurship

ECTS: 5 credits

Course level: Intermediate

Course description

The aim of this course is help students launch their businesses. All along the term, students will be given several workshops, run jointly by emlyon and Centrale Lyon's incubators, to learn various aspects of entrepreneurship (from ideation all the way to business model), between which they will work on their businesses. Students will be welcome to work alone, to group around common ideas, or join existing projects from emlyon and Centrale Lyon's incubators.

Assessment tools: presentations and pitch

Prerequisite: None

Academic and Professional Objectives

Section	Learning Outcomes
6.1.1 Concepts and Theories Introduced in the Course	1. Entrepreneurship 2. Field studies 3. Business model
6.1.2 Knowledge Acquired During the Course	1. Know the steps towards opening a business 2. Understand field studies for adapting a business idea 3. Understand brand, marketing, and sales aspects of a business 4. Understand corporate finance to run projections 5. Know legal aspects of launching and running a business
6.1.3 Competencies Developed During the Course	1. Identify a business potential 2. Measure value creation 3. Conduct field studies to fine tune a business idea 4. Run financial projections and build a business model 5. Choose the right legal structure for launching a business

If you have any questions, please contact us at the following address:

incoming.globalbba@em-lyon.com

we are makers

em lyon business school